GPU bench power box

Build guide: wiring schematic, parts list and directions. Revision 1, 20 September 2026.

This box sits between a current-limited bench supply and a graphics card lying on the bench. It powers the card through a PCIe riser and its aux connectors with no motherboard, shows what the whole card draws, and gives you three injection outputs on the same ground. It is modelled on the testers shown by northwestrepair (2023) and GPU Solutions (2026), with four changes: separate switches for the riser and the aux connectors, individually switched and fused 3.3 V / 1.8 V rails, a backup fuse, and an off-the-shelf 8-pin to 16-pin adapter for 12VHPWR / 12V-2x6 cards.

The box has no current limiting of its own. The bench supply's CC limit is the protection. Never feed the box from an ATX or server supply, or anything else that can deliver tens of amps into a shorted card. The 3.3 V and 1.8 V outputs can damage a logic pin or back-power a GPU: keep their switches off until the moment you need them.

What it tells you

Reading at 12.5 V	Usual meaning
About 0 A	Rails are not coming up. Check 3.3 V, 5 V, 1.8 V and the enable chain.
Roughly 1–2.5 A	Card is idling. About 1.1–1.4 A was shown for an RX 7800 XT and about 2 A for an RTX 3080 / 3090.
Supply drops into CC	A short or a stuck phase. Switch off, then find the short with the variable output and the thermal camera.
SW1 only vs SW2 only	Tells you whether the draw is on the slot 12 V input or the aux input.

Treat the idle figures as starting points from the two videos. Build your own table from known-good cards in section E.

Supply settings

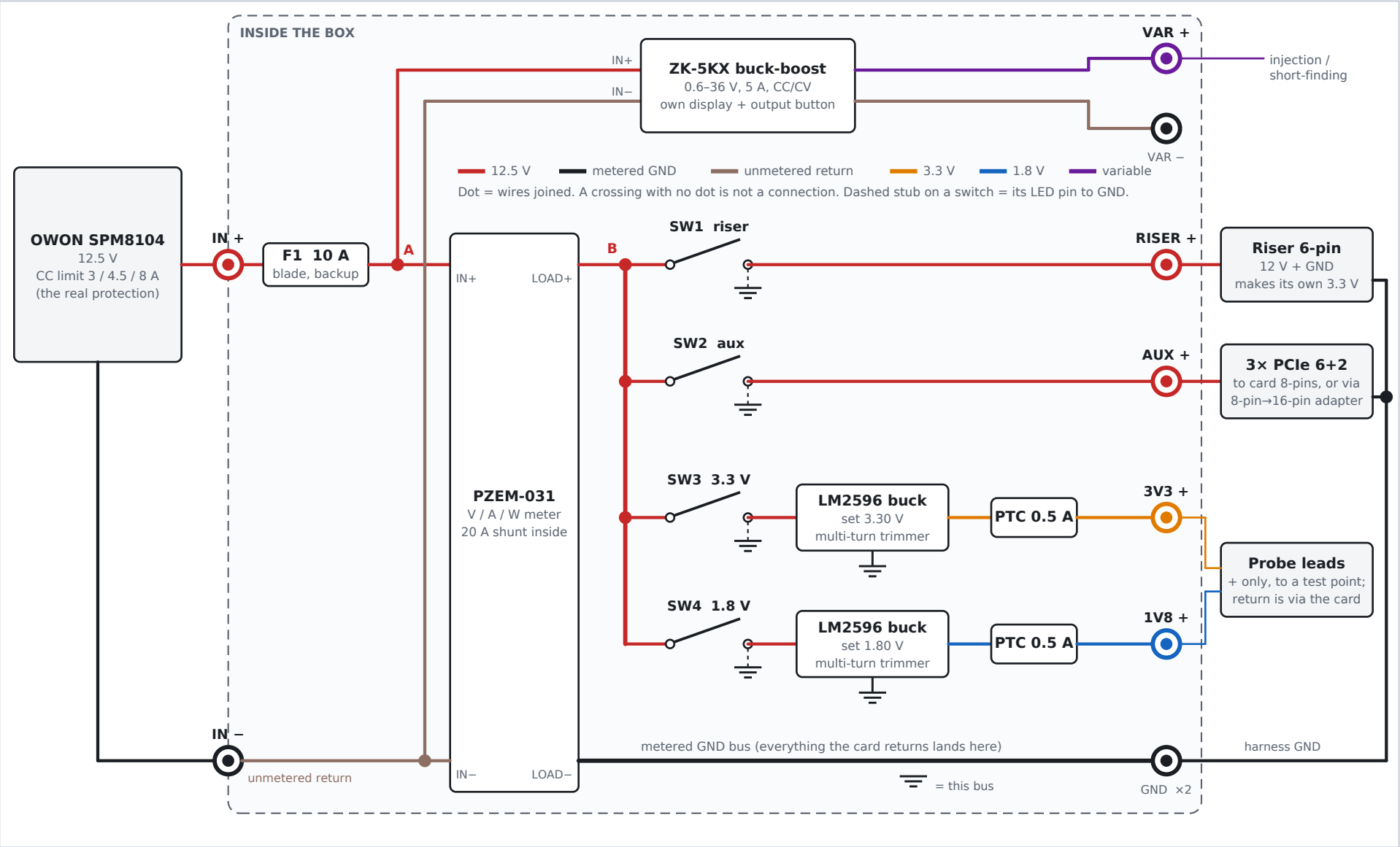
Card	Voltage	Current limit
Anything up to RTX 4090	12.5 V	3 A
RTX 5080	12.5 V	4.5 A
RTX 5090	12.5 V	6–8 A

12.5 V rather than 12.0 V covers the drop through the fuse, meter shunt, switch and leads. Set the limit first, confirm it, then enable the output.

Known unknowns

- **PERST#.** On a bare riser the reset line floats. The cards in both videos sequenced their rails anyway, but it is not guaranteed for every generation. If a known-good card stays near 0 A, that is the first thing to look at.
- **PZEM shunt position.** I could not confirm whether the shunt is in the positive or negative leg. Wiring it as drawn is correct either way.
- **This box proves a card idles safely and nothing more.** A card that passes still needs a test PC and memory tests before it goes back to a customer.

Wiring schematic



The PZEM-031 is drawn as a four-terminal block because that is how it wires: supply on one pair, load on the other. Follow the diagram printed on the meter. Everything the card powers or returns through must connect on the LOAD side, or the meter will not see it. The ZK-5KX is fed from before the meter on purpose, so injection current does not disturb the card reading. When a card is powered from the box, use only the + lead of an injection output and let the current return through the card's ground.

Parts list

#	Part	Qty	Check before buying
1	Peacefair PZEM-031 DC meter	1	6.5–100 V with the built-in 20 A shunt. Not the PZEM-051, which uses an external shunt.
2	ZK-5KX buck-boost CC/CV module	1	0.6–36 V, 5 A, display and output button.
3	LM2596 buck modules	2	Blue multi-turn trimmer. Sold in 10-packs.
4	PCIe x1→x16 mining riser	1	VER009S or VER010-X with a 6-pin power input. Only the slot board is used.
5	4 mm banana jacks, panel mount	10	15 A or better. Red and black; mark the others with coloured rings or labels.
6	4 mm banana plugs + stackable leads	4 + 1 pair	14–16 AWG leads for supply to box.
7	12 V DC illuminated rocker switches	4	Rated at 12 V DC, 16–20 A, 20 mm round, 3-pin.
8	Inline ATC blade fuse holder + 10 A fuses	1	14–16 AWG leads.
9	Radial PTC resettable fuses, 0.5 A hold	2	From an assortment kit.
10	PCIe 8-pin female → dual 6+2 male splitter	2	18 AWG. Cut off the female end: four 6+2 plugs.
11	8-pin PCIe → 16-pin 12VHPWR / 12V-2x6 adapter	1	2× or 3× 8-pin inputs, the style shipped with RTX 40 / 50 cards. It sets the sense pins.
12	Silicone wire: 16 AWG red/black, 22 AWG	1 each	16 AWG on every 12 V and ground run.
13	Wago 221 lever nuts, assorted	1 kit	Genuine 221 series, or ferrules.
14	Banana to mini-grabber / needle-probe leads	1 set	For the injection outputs.
15	M3 screws and heat-set inserts	1 kit	For the printed enclosure.

Links go to listings that existed on 20 September 2026; the rest are generic parts to search for. Rows 1, 4, 10 and 12 plus two switches and five jacks are enough for a first power-up.

Build directions

A. Test every module on its own

1. Set the bench supply to 12.5 V with a 0.5 A limit for all module tests.
2. LM2596 #1: power it with nothing on the output and trim to 3.30 V. Repeat for #2 at 1.80 V.
3. Check both bucks for power-on overshoot: watch the output on a scope, or a DMM on min/max, while switching the input on and off several times. Reject a module that spikes above its setpoint.
4. ZK-5KX: set 1.0 V and a 0.5 A limit, enable its output, short the output leads briefly and confirm it holds 0.5 A in CC.
5. Riser: feed 12.5 V into its 6-pin with no card. Find the 12 V and ground pins with a DMM, and confirm 3.3 V appears on slot pins A9, A10 and B8. Confirm those pin numbers against the PCIe CEM pinout before relying on them.

B. Prepare the harnesses

6. Cut the female end off both splitters. In every 6+2 plug the three 12 V wires are usually yellow and the rest are ground, including both sense pins. Confirm with a continuity check against a PCIe 8-pin pinout, not by colour alone.
7. Aux harness: join the 12 V wires of three 6+2 plugs to one red banana plug and their grounds to one black banana plug.

8 Riser harness: the fourth 6+2 plug, used as a 6-pin, to its own red and black banana plugs.

9 Do not modify the 8-pin to 16-pin adapter. It plugs onto the aux harness when a card needs it.

C. Wire the box

10 Print or drill the enclosure for ten jacks, four switches, the PZEM-031 and the ZK-5KX. Label every jack.

11 IN + jack to the fuse holder, fuse holder to node A. Use a lever nut for A.

12 Node A to PZEM IN+ and to ZK-5KX IN+. IN − jack to PZEM IN− and to ZK-5KX IN−.

13 PZEM LOAD+ to node B. From B run four wires, one to each switch input.

14 PZEM LOAD− to the ground bus. Land on it: both GND jacks, both LM2596 IN−, and the LED pin of every switch.

15 SW1 output to RISER +. SW2 output to AUX +.

16 SW3 output to LM2596 #1 IN+. Its OUT+ through a 0.5 A PTC to the 3V3 jack. Same for SW4, LM2596 #2 and the 1V8 jack.

17 ZK-5KX OUT+ to VAR +, OUT− to VAR −.

18 Use 16 AWG for every 12 V and ground run and 22 AWG for the two injection rails.

D. Check it before a card goes near it

19 Power off: check continuity from IN + to each output with its switch on and none with it off. Check for no continuity between any + jack and ground.

20 Power on at 12.5 V / 0.5 A, no card. The PZEM should read 12.5 V and close to 0 A. Verify with a DMM: 12.5 V at RISER + and AUX +, 3.30 V and 1.80 V at their jacks, each only while its switch is on.

21 Plug in the harnesses and measure at the far end: 12 V on the right pins of every 6+2 plug and the riser 6-pin, ground on the rest.

E. First card

22 Use a known-good card. Set 12.5 V and a 3 A limit. All four switches off. Seat the card in the riser and connect the aux plugs.

23 SW1 on, then SW2 on. Note the PZEM current and compare it with the supply's own reading. Record it as that model's baseline. Switch off and repeat with each known-good card you have.

24 Try the same card with SW1 only and SW2 only, and record both. Some cards will not sequence without both inputs; that is worth knowing per model.

Sources

- northwestrepair, Introducing Super Mega Tester Pro XL (2023)
- GPU Solutions, How to Build a Custom GPU Testing Power Supply (2026)
- GPU Solutions, GPU Repair Power Supply Control Unit (2026)
- GPU Solutions build guide, circuit diagram and print files
- 12VHPWR / 12V-2x6 sense-pin table